

Rubric:

Use the example calculation worked out in the text to come up with an estimate of the time required for half the air to leak out of the International Space Station through the hole shown in the photograph in the Guardian article (reference above).

- a. Go to the official NASA website for Facts and Figures for the International Space Station and find the “total pressurized volume”, V , of the ISS. Express the value in cubic meters.

- (4 pts): The total pressurized volume is 35,491 cubic feet which converts to 1005 cubic meters
- -1 pt for incorrect conversion
- -1 pt for incorrect number

- b. Consult the recommended web articles (the Guardian and the official NASA news) to find an estimate of the diameter of the hole that caused the air leak. (Note: The diameter will probably be given in units of millimeters. Be sure to convert this to meters. Calculate the area of the hole, A , in units of square meters.)

- (6 pts): A hole that is 2 millimeters in diameter was found; it's area in square meters is 3.142×10^{-6} square meters.
- -1 pt for incorrect number
- -1 pt for incorrect version
- -1 pts for incorrect area calculation

- c. Using the example calculation in the text (and shown in class) estimate the time required for half of the air to leak out of the ISS.

- (10 pts): Use the equation $\Delta t = \frac{\Delta n}{n} \frac{6V}{A} \sqrt{\frac{m}{3k_B T}}$, where $\Delta n/n = 1/2$, $V = 1005 \text{ m}^3$, $A = 3.142 \times 10^{-6} \text{ m}^2$, $T = 300 \text{ K}$, $k_B = 1.38 \times 10^{-23} \text{ m}^2 \text{ kg s}^{-2} \text{ K}^{-1}$.

The mass per particle should be calculated for something close to N_2 such that $m_{\text{N}_2} = 2 \times 14 \times 1.66054 \times 10^{-27} \text{ kg} = 4.65 \times 10^{-26} \text{ kg}$.

The answer should then be close to $\Delta t_N = 1.86 \times 10^6 \text{ s} = 21.53 \text{ days}$

- -2 pts for incorrect formula application
- -1 pt each for incorrect numbers for each variable (up to 3 pts)
- -1 pt for incorrect answer